

- Can you find another set of rules that create Kurushima's squares? For example, are there rules that enable one to travel up and down through Kurushima's square instead of diagonally?
- Are there other types of magic squares that satisfy the definition of a magic square but don't follow Kurushima's rules at all?
- Is there a more efficient way to write code to implement Kurushima's algorithm?

Magic squares occupied the attention of great Japanese mathematicians for several centuries, and they've found a significant place in cultures around the world. We can count ourselves lucky that the great mathematicians of the past gave us algorithms for generating and analyzing magic squares that we can easily implement on today's powerful computers. At the same time, we can admire the patience and insight that was required for them to investigate magic squares with only pen, paper, and their wits (and the occasional magical turtle) to guide them.

Summary

In this chapter, we discussed some historical algorithms that range from a few centuries to a few millenia old. Readers who are interested in historical algorithms can find many more to study. These algorithms may not be of great practical utility today, but it can be worthwhile to study them—first because they give us a sense of history, and second because they help broaden our horizons and may provide the inspiration for writing our own innovative algorithms.

The algorithms in the next chapter enable us to do some commonly needed and useful tasks with mathematical functions: maximize and minimize them. Now that we have discussed algorithms in general and algorithms in history, you should be comfortable with what an algorithm is and how one works, and you should be ready to dive into serious algorithms used in the most cutting-edge software being developed today.